



Introduction

Foreword

Intended use of the machine

Only use the machine for the products and parameters for which it was designed.

The only changes of the machine that are permitted are given in this manual. For all other changes you must obtain the written approval of **BluePrint Automation**.

Personnel

Only trained personnel are permitted to operate the machine or packaging line. Operators must sufficiently understand the content of the manual.

Only trained personnel are permitted to do maintenance on the machine or packaging line. They must sufficiently understand the content of the manual.

All personnel must carefully read the chapters that are related to their tasks, refer to chapter 1. All personnel and persons that work near the machine must read and understand the Safety Chapter of this manual. If you have questions always contact **BluePrint Automation**.

Purpose Of The Manual

This manual contains descriptions, procedures and tips on how to operate and do maintenance on the machine safely and efficiently.

This manual is part of a set of documentation. It can be a digital format, a hard copy, or both.

Make sure the manual and related documents are always available for the personnel that work with or on the machine.

Illustrations

This manual contains some general illustrations, drawings, and photos. These may sometimes be different from the machine that you purchased.

Integrated Modules

If third party modules are integrated into the machine, the instructions are not included in this manual, but supplied under separate cover. For these modules we will refer to the OEM (Original Equipment Manufacturer) documentation.

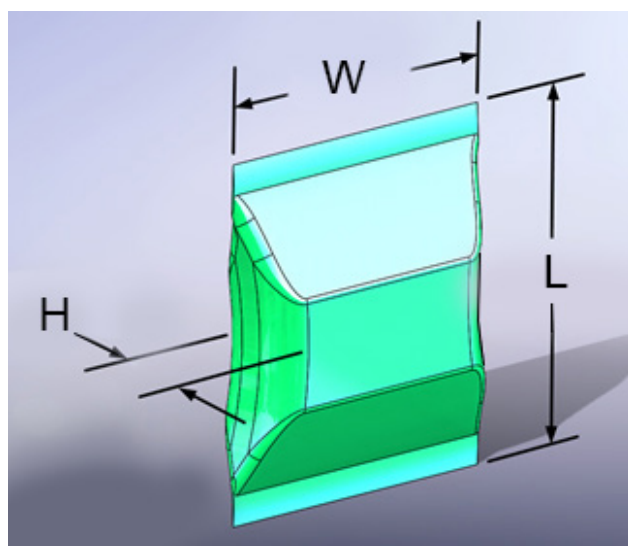
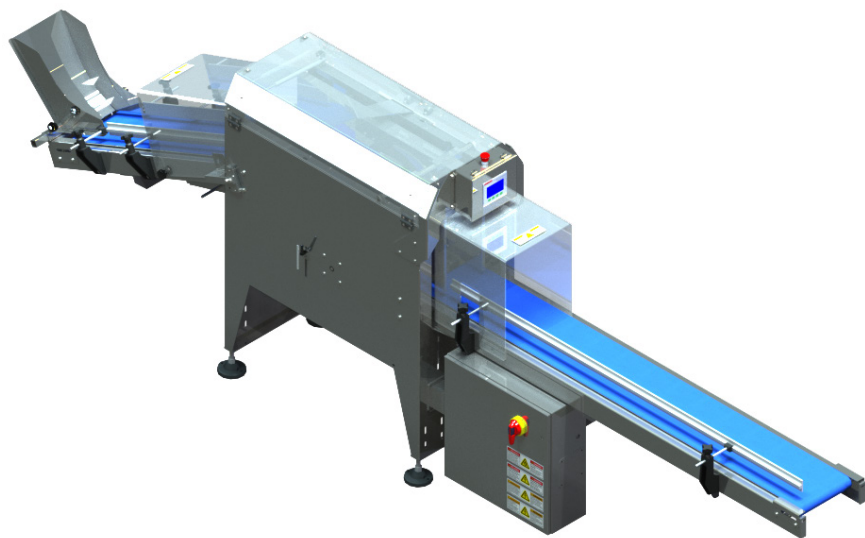
Typography

In this manual different typography is used for specific intent. The next table shows the different fonts and their meanings.

Example Text	Meaning
System Start	The text on a hardware push-button or HMI touch-cell.
<i>Electrical Schematics</i>	The name of a document included in the documentation package.
1. First step. a. First sub-step. b. Second sub-step. 2. Second step.	All sequential procedures have numbered steps. Do each step in the given sequence. Sometimes a step has one or more sub-steps. Do each sub-step in the given sequence.
• Item. • Item. • Item.	Each item in a list has a bullet in front of it. The sequence of the items in a list is not important.
Boldface.	Used for emphasis.

Description

The **Collator 200** is designed to receive packages from upstream equipment and reorient them such that they are arranged vertically collated front-to-back in a single row on the exit conveyor so that an operator can pick them up and place them into cases.



Standard Collator 200 Machine Limits:			
Parameter	Units	Minimum	Maximum
Package Width (W)	in [mm]	4 [101,6]	12,5 [317,5]
Package Length (L)	in [mm]	4 [101,6]	17,25 [38,15]
Package Thickness (T)	in [mm]	0,75 [19,05]	5,25 [133,35]
Weight	lb [kg]	0,06 [0,027]	5 [2,268]
Rate	package/min	-	130
Cut off height of bagger	in [mm]	33,5 [850,9]	38 [965,2]

Glossary

The following lists and defines commonly used abbreviations, acronyms, and initialisms in the packaging machinery industry.

ARB – Activated Roller Belt conveyor: On the ARB™ Intralox conveyor, products rest on free-spinning angled rollers rather than on the belt surface. These rollers extend above and below the belt surface and are positioned at an angle in relation to the direction the belt travels. Rollers that are activated by the carry-way surface below move products across the direction of roller orientation rather than the direction of driver travel, thus making it possible to move conveyed products selectively. ARB conveyor equipped machinery can change the direction, alignment, location, and speed of an item independently without using rails or complicated mechanical controls.

BCID – Bar Coded Identification Markings: A bar-code is an optical, machine-readable representation of data; the data usually describes something about the object that carries the bar-code. Traditional bar-codes systematically represent data by varying the widths and spacings of parallel lines, and may be referred to as linear or one-dimensional.

CSSC – Centre Special Slotted Case: The inner and outer flaps are different lengths. Both pairs of flaps meet at the centre of the box. The inner flaps meet in so as to provide a level rest for the contents of the box.

DCC – Double Cover Case: Frequently used for tall or heavy products that would be difficult to lower into a box. The product is placed on the bottom cover, and the centre is placed over the product. The two interchangeable covers (top and bottom) of the DC box are tray style and usually extend only a few inches down the centre body.

EOAT – End of Arm Tooling: Or end effector is the device at the end of a robotic arm, designed to interact with the environment.

FLA – Full Load Amperage: the machine's current draw under full load in amperes (A).

FOL – The Full Overlap Case: The FOL configuration is ideal for boxes with relatively narrow widths. The outer flaps overlap the full width of the box making it especially resistant to rough handling. And, the extra thicknesses of corrugated material on the top and bottom panels provide added stacking strength.

HDP – Horizontal Delta Packer: A delta robot system that loads cases with product arranged horizontally in the case.

HMI – Human Machine Interface: Or System Control Interface is a touch-screen that provides the operator with an interface to start/stop the work-cell and to select new products. Maintenance personnel use the screen for system maintenance, parameter setting, and troubleshooting. The HMI is located on the door of the Robot Control Enclosure.

HPP – Horizontal Pick and Place: Robotic gantry that loads cases with product arranged horizontally in the case.

HSC – Half-Slotted Box with Cover: The HSC is frequently used for both shipping and shelf storage, and in applications where the cover must be removed and replaced frequently. The top of the HSC box is a tray style which extends partially (usually less than 2/3 the depth of the bottom) PHSC, but can be made to extend the full depth of the bottom FHSC if extra strength is needed.

HVDP – Horizontal/Vertical Delta Packer: A delta robot system that loads cases with product arranged either horizontally or vertically in the case dependent on the operational set-points and change parts installed.

HVPP – Horizontal/Vertical Pick and Place: Robotic gantry that loads cases with product arranged either horizontally or vertically in the case dependent on the operational set-points and change parts installed.

ICE – Integrated Case Erector: Integrated robotic case erector.

KD – Knock Down: Corrugated cases fed to the integrated case erectors.

MCP – Main Control Panel: Enclosure that houses the system controls such as conveyor VFDs, PLC, power distribution devices, etc.

MGC – Modular Gate Case Packer: A case packer that loads the cases by dropping (gravity) the product into the case by means of a modular gate.

MPD – Missing Product Detector: A piece of machinery designed to weigh a case after the packing process in order to ascertain that the specified product quantity has been placed into the case; cases that are out of specification can then be rejected.

OSC – Overlap Slotted Case: style of box is similar to the RSC box, except the outer flaps overlap by an inch or more. The OSC style of box is used when the length of the box is considerably greater than the width and results in an excessively long distance between the end flaps. This makes it desirable to overlap the outer flaps to prevent them from spreading apart. The OSC box is usually closed with staples through the overlapping flaps.

PLC – Programmable Logic Controller: Or programmable controller is an industrial digital computer which has been ruggedised and adapted for the control of manufacturing processes, such as assembly lines, or robotic devices, or any activity that requires high reliability control and ease of programming and process fault diagnosis.

POL – Partial Overlap Case: With this box, all flaps have the same length. The outer flaps overlap by 1 inch or more.

PPE - Personal protective equipment: PPE refers to protective clothing, helmets, goggles, or other garments or equipment designed to protect the wearer's body from injury or infection. The hazards addressed by protective equipment include physical, electrical, heat, chemicals, bio-hazards, and airborne particulate matter.

RFID - Radio Frequency Identification: A wireless system comprised of two components: tags and readers. The reader is a device that has one or more antennæ that emit radio waves and receive signals back from the tag.

RGP – Rotary Gate Packer: A case packer that loads the cases by dropping (gravity) the product into the case by means of a rotary gate.

RSC – The Regular Slotted Case: is the most common box style used for shipping and storage. The two outer flaps of an RSC box meet at the centre of the box when folded and are usually closed using tape, or staples.

RSD – Regular Slotted Display: This case is carefully designed, with the retailer in mind. Packing and closing is similar as an RSC case. There is no need to tear the front, as this case has a pre-cut front flap. This makes it easy to open, and shelf ready in mere seconds.

SCCR – Short Circuit Current Rating: a rating on components and assemblies representing the maximum level of short-circuit current that a component or assembly can withstand. This rating is used to determine compliance with NEC 110.10. It is measured in kilovolt-amperes (**kVA**).

SKU – Stock Keeping Unit: In the field of inventory management, a stock keeping unit is a distinct type of item for sale, such as a product or service, and all attributes associated with the item type that distinguish it from other item types. For a product, these attributes could include manufacturer, description, material, size, colour, packaging, and warranty terms. When a business takes inventory of its stock, it counts the quantity it has of each SKU. SKU can also refer to a unique identifier or code that refers to the particular stock keeping unit.

VDP – Vertical Delta Packer: A delta robot system that loads cases with product arranged vertically in the case.

VFD – Variable Frequency Drive: A variable-frequency drive (also termed adjustable-frequency drive, variable-speed drive, AC drive, micro drive or inverter drive) is a type of adjustable-speed drive used in electromechanical drive systems to control AC motor speed and torque by varying motor input frequency and voltage.

VFFS – Vertical Form Fill and Seal machine: Vertical automatic packing machines using a single film coil to obtain a package with three seals: two transverse seals and one longitudinal seal. Its main field of application includes loose products, granulated products, doughy products or products difficult to handle. This type of machine is aimed at both food and non food markets.

VHPF – Vertical/Horizontal Pack Former: A case packing system that can load cases with product arranged either vertically or horizontally depending on installed change parts and operating parameter settings.

VPF – Vertical Pack Former: Case packing system that loads cases with product arranged vertically in the case.

VPP – Vertical Pick and Place: A robotic gantry system that loads cases with product arranged horizontally in the case.